Wireless Security Lab: Smartphones Spoofing

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ABSTRACT

This laboratory evaluates GNSS spoofing effects on Android smartphones across open-space, RF-interference, and urban multipath environments using a software-injected attacks simulating coordinate displacement and time delays.

1 METHODOLOGY AND EXPERIMENTAL DESIGN

1.1 Data Collection Infrastructure

1.1.1 Hardware Specifications. The experimental campaign utilized three distinct Android smartphones to ensure diversity in chipset architecture and GNSS firmware behavior.

- **Smartphone A**: Asus ZenFone 4 Max; OS: Android 8.1.0; Chipset: Qualcomm Snapdragon 430.
- **Smartphone B**: Xiaomi Redmi Note 10; OS: Android 12; Chipset: MediaTek Helio G95.
- **Smartphone C**: OnePlus Nord 2T; OS: Android 14; Chipset: MediaTek Dimensity 1300.

1.1.2 Software Compatibility and Versioning. All datasets were logged using `GNSSLoggerv3.1.1.2`, available from the official Google repository.

1.2 Experimental Environments and Procedures

1.2.1 Environmental Baseline: OpenSpace Dataset. A reference baseline under nominal GNSS conditions established using Smartphone A in the area above Politecnico Rooms I.[1] This location provides an unobstructed hemispherical view of the sky with optimal satellite geometry and minimal signal attenuation. The resulting data serve as a control group for evaluating the degradation effects observed in more complex urban or high-interference scenarios.

1.2.2 Radio-Frequency Interference: Maddalena Dataset. Smartphone B was deployed at Colle della Maddalena[2], where dense broadcast infrastructure served to evaluate if adjacent-band emissions desensitize the GNSS receiver's low-noise amplifier (LNA) or elevate the noise floor, as reflected in C/N_0 distributions.

1.2.3 Multipath and Obstruction: Dumarey Building Dataset. To analyze near-field obstruction and Non-Line-of-Sight (NLoS) impacts, Smartphone C logged static, ground-level GNSS data at the Dumarey facility[3] (Politecnico di Torino). Positioned near the facade, this proximity physically obstructed a vast sector of the sky dome, causing severe azimuthal masking and inflating GDOP. This structure forced NLoS signals to bounce, strongly emphasizing multipath phenomena and inducing significant geometric delays that likely drove the inflated pseudorange variance and systematic biases subsequently observed in the `GpsWlsPvt` estimation.

1.2.4 Power Management Constraints. A fourth measurement was deployed using Smartphone B with power saving enabled to investigate on modifications in `HardwareClockDiscontinuityCount`. There were no immediate discontinuities, suggesting that Smartphone B's chipset may resist clock duty-cycling during these specific logging durations, though further investigation is required for generalization.

1.3 Analytical Processing Pipeline

The processing pipeline employs a modified version of van Diggelen's open-source *GPS Measurement Tools* [4], extended by the NavSAS Research Group with spoofing simulation capabilities. The workflow proceeds as follows: `ReadGnssLogger` parses raw logs applying filters; `GetNasaHourlyEphemeris` retrieves precise ephemerides from NASA CDDIS; `ProcessGnssMeas` converts raw counts to pseudoranges, integrating synthetic ranges via `compute_spoofSatRanges` when `spoof.active=1`; and `GpsWlsPvt` computes the Weighted Least Squares position-velocity-time solution. The pipeline generates six diagnostic outputs (pseudoranges, C/N_0, PVT trajectories, residuals).

1.4 Spoofing Simulation Parameters

Cyberspoofing was emulated via MATLAB manipulation of `spoof.position` and `spoof.delay`. Authentic pseudoranges were replaced with geographically plausible yet statistically significant counterfeit values. A 15-second pre-attack window facilitated comparative analysis between the stable baseline and induced instability. The applied `spoof.delay` (1×10^{-3} s) emulated meaconing-induced propagation latency, evaluating WLS clock bias resilience against uniform pseudorange inflation.

2 RESULTS AND DISCUSSION

2.1 MATLAB Plot Outputs

`Pseudorange Change from Initial Value` shows range variations normalized to zero, producing a "scissor" pattern as satellites approach or recede. `Hardware Clock Discontinuity` detects clock resets from power management that introduce pseudorange step-functions. `Pseudorange Rate (diff(PR)/diff(time))` shows the discrete-time derivative of pseudoranges; abrupt variations indicate spoofing or tracking anomalies. `C/N0` displays signal power for each satellite; concurrent attenuation across channels indicates RF interference. `Raw Pseudoranges` plots horizontal position with a circle centered at the median containing 50% of positions (CEP radius). `Horizontal Speed` estimates velocity magnitude; large deviations from zero (for static receivers) indicate outliers. `HDOP`

[1] https://maps.app.goo.gl/3r5E822u71dQwHqw7
[2] https://maps.app.goo.gl/6ASxJNceVYFJbe8x6
[3] https://maps.app.goo.gl/si8SUPk2TyUSSXdZA

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quantifies geometric dilution; lower values indicate better accuracy. The orange trace shows satellite count. `Common Bias Clock Offset` shows receiver clock bias; linear drift indicates oscillator instability.

2.2 Dataset 1 - Open Space

2.2.1 Data Presentation. The "Pseudoranges vs. time" plot (Figure 1) demonstrates continuous tracking of all visible satellites throughout the 180-second acquisition. The C/N_0 analysis (Figure 3) shows stable signal quality with a mean of 29.76 dB-Hz. PRN 16 maintains the highest strength at approximately 38 dB-Hz, while lower-elevation satellites exhibit reduced but stable values (22–33 dB-Hz). No signal attenuation or tracking instabilities are observed. The pseudorange rate plot (Figure 2) reveals consistently negative derivatives (-1200 to -2000 m/s) for all satellites. Zero hardware clock discontinuities confirm power-saving modes were disabled. The WLS position solution (Figure 4) yields median [45.0651389°, 7.6584375°, 313.79 m] with 50% CEP of 4.25 m and maximum horizontal error of 16.01 m. Altitude varies from 262.26 m to 366.19 m (103.93 m range, 20.10 m standard deviation). Horizontal distribution shows anisotropic dispersion: East Variance (27.20 m²) exceeds North Variance (4.50 m²) by factor of six. Satellite geometry remains stable with 4–6 tracked satellites throughout the measurement period. Horizontal velocity (Figure 4) oscillates near zero with occasional ±10 m/s spikes. Common bias clock offset exhibits cumulative linear drift of -3×10^5 m (-300 km equivalent range) over 180 seconds, corresponding to a drift rate of -1593.92 m/s.

2.2.2 Interpretation. The stable C/N_0 profiles and continuous tracking confirm this unobstructed environment provides an effective reference baseline, validating nominal receiver performance under optimal conditions. The systematic negative pseudorange rates indicate uncorrected hardware clock drift rather than genuine satellite convergence. Since orbital mechanics precludes simultaneous approach of all satellites to a static receiver, this uniform shift reflects mass-market oscillator instability. The WLS filter absorbs this common-mode timing error into the clock bias state, evidenced by the linear -1593.92 m/s drift rate, confirming a receiver-side artifact. The tight 4.25 m CEP demonstrates achievable precision under open-sky conditions. However, East-West anisotropy indicates residual geometric dilution along the longitudinal axis. The altitude variability (103.93 m range) reflects typical vertical dilution in consumer-grade GNSS, where geometric constraints degrade altitude accuracy relative to horizontal positioning. Velocity oscillations near zero with occasional spikes represent static receiver noise amplified by discrete-time differentiation, not actual motion.

2.2.3 Limitations. The primary limitation is the systematic hardware clock drift affecting all pseudorange rates. While the WLS algorithm successfully absorbs this common-mode error into the clock bias state, the artifact obscures the true radial velocity signatures of individual satellites, preventing independent validation of Doppler-based velocity estimation. The East-West positional anisotropy, while moderate compared to constrained environments, indicates that even under open-sky conditions, satellite constellation geometry introduces directional bias. This specific geometric characteristics may not generalize to other times or locations.

2.3 Dataset 2 – Maddalena

2.3.1 Data Presentation. Visual inspection of the observables confirms the presence of significant RF interference. The "Pseudoranges vs. time" plot (see Figure 5) demonstrates the receiver's inability to maintain continuous phase lock on several signals, resulting in discontinuous tracking trajectories. Despite these intermittent tracking anomalies, the "Pseudoranges change from initial value" plot preserves the characteristic "scissor" signature, where the diverging and converging pattern accurately reflects relative orbital kinematics as satellites approach and recede.

The Weighted Least Squares (WLS) position solution metrics (see Figure 7) reveal that the number of satellites utilized for PVT computation exhibits notable fluctuations, periodically degrading from a peak of 9 down to 6 during epochs of acute interference, while sustaining an average of 8 satellites. The system maintains HDOP values predominantly bounded between 1.5 and 2.5, indicating adequate spatial geometry was preserved despite hostile RF conditions.

The C/N_0 analysis (see Figure 6) highlights five distinct temporal epochs characterized by abrupt, concurrent signal power attenuation across multiple channels. Precisely during these five epochs of severe C/N_0 degradation, the receiver struggles to demodulate incident signals, consequently dropping 2 to 3 satellites from the position fix.

Subsequently, a simulated location spoofing attack was injected with target coordinates at approximately 100 m spatial offset from the true collection site (spoof.position = [45.0313399, 7.7213363, 776.91]), algorithmically initiated at $t = 15$ s. The spatial trajectory is abruptly translated to the counterfeit coordinates (see Figures 8 and 9). The 15-second delay results in initial samples correctly depicting the true location, whereas all subsequent epochs strictly conform to the spoofed trajectory. In the pseudorange rate domain, location spoofing induces severe perturbations, transitioning from almost-linear nominal behavior to highly erratic patterns. The pseudorange rates, computed as discrete time derivative $\frac{\Delta\rho}{\Delta t}$, exhibit extreme irregularities due to the instantaneous spatial translation commanded by spoofing parameters.

2.3.2 Interpretation. The tracking instability and intermittent signal dropout are directly attributable to electromagnetic interference from the adjacent high-power radio broadcasting antenna. The correlation between C/N_0 degradation epochs and satellite dropout confirms that RF interference desensitizes the receiver's front-end, elevating the noise floor and disrupting tracking loops. However, the preserved "scissor" pattern in pseudorange changes demonstrates that orbital kinematics remain correctly captured during brief periods of stable lock.

The maintained HDOP range (1.5–2.5) despite satellite losses indicates that the remaining constellation geometry remained geometrically favorable. This resilience allowed the receiver to sustain positioning capability even under hostile RF conditions.

The catastrophic pseudorange rate perturbations following spoofing injection are a direct mathematical consequence of the discontinuous spatial translation. The discrete-time derivative amplifies the instantaneous coordinate jump into extreme numerical discontinuities, providing an unambiguous attack signature.

2.3.3 Limitations. The primary limitation is reliance on post-processing software injection rather than physical over-the-air RF spoofing. While this computational method effectively alters mathematical observables and validates processing tool response, it does not replicate hardware-level receiver dynamics such as Automatic Gain Control (AGC) variations or authentic correlation peak distortions. Additionally, the exact transmission power and spectral characteristics of the interfering broadcasting antenna were not quantitatively measured, limiting the ability to establish rigorous mathematical correlation between precise interference power and observed C/N_0 degradation.

2.4 Dataset 3 – Dumarey

2.4.1 Data Presentation. Initial baseline signal analysis reveals a significant degradation in overall quality. Specifically, the C/N_0 of PRN 15 exhibits a highly periodic fluctuation (17–27 dB-Hz) before repeatedly losing lock. While the exact physical mechanism remains undetermined, this sawtooth pattern strongly suggests severe multipath fading, where constructive and destructive reflections cyclically disrupt the receiver's tracking loops. Regardless of the exact origin, this signal instability directly correlates with cyclic variations in the estimated altitude (see Figure 10). Consequently, the altitude produced a massive 321.88 m range (Table 1, Baseline), shifting the median to 476.71 m—a severe vertical bias when compared to the 249 m true topographical altitude.

Furthermore, HDOP metrics highlight poor satellite geometry (fluctuating between 1.5 and 3.5), exhibiting perfect inverse proportionality to the number of tracked satellites, which averaged only 5.5. As a result, 50% of computed positions fall within a broad 44.6 m CEP radius (Table 1, Baseline), centered at [45.063668, 7.658522] (see Figure 11). This dispersion is heavily driven by spatial outliers reaching a maximum distance from the mean of 220.79 m. Visually, the spatial distribution forms a straight line strictly perpendicular to the facade. This geometric distortion is mathematically confirmed by the directional variances: the East Variance is an order of magnitude larger than the North Variance, as shown in Table 1. The underlying cause of this distribution is discussed below.

A programmatic data filter was implemented to exclude observations below 20 dB-Hz and remove erratic satellites (namely PRNs 2, 15, and 16). Eliminating these noise-dominated signals successfully stabilized the WLS altitude, reducing its error range to 42.69 m and eradicating the periodic artifacts. In this filtered solution, the 50% CEP collapsed to a tightly constrained 4.48 m radius (Table 1, Filtered), representing a full order of magnitude reduction compared to the baseline (see Figure 12).

Subsequently, a delay spoofing attack was emulated on the filtered dataset. The attack shifted the estimated median to [45.0639782, 7.6586818, 301.951] (see Figure 13) and drastically expanded the 50% CEP to 122.22 m (Table 1, Spoofed). Although the linear distribution persisted, outliers reached a 554.40 m maximum horizontal error. Moreover, latitude and longitude deviations spiked to roughly ±500 m, while altitude variations exploded to a 3200.36 m range (Table 1, Spoofed). Despite this severe spatial disruption, the attack's onset remains trivially detectable via the PRR derivatives, which mathematically diverge precisely at the 15-second mark.

2.4.2 Interpretation. The data robustly illustrates GNSS vulnerability in constrained environments. The orthogonal, straight-line dispersion of the baseline positions is the direct result of poor cross-track geometry combined with severe multipath. With half the sky obscured, the receiver completely lacks transversal spatial diversity. Therefore, it is highly probable that the Dumarey facade reflected NLoS signals, artificially lengthening their pseudoranges and forcing the WLS estimator to stretch the solution along the single, geometrically constrained East-West axis.

Under the spoofing attack, the catastrophic PVT degradation highlights the system's critical sensitivity to timing manipulation. A 1 ms artificial delay injected into the tracking loops translates instantaneously into a 300 km synthetic range extension. Without RAIM to isolate the outlier, this discontinuous pseudorange jump forces the navigation filter to diverge, completely hijacking the PVT solution.

Additionally, a minor, global C/N_0 attenuation observed at the log's end likely resulted from human-body path loss. This perturbation occurred inadvertently when the operator physically leaned over the smartphone's internal antenna.

2.4.3 Limitations. A significant anomaly in the PRR observables showed only satellites presenting a negative slope. While the varying negative gradients correctly imply different radial velocities, it is highly improbable from an orbital mechanics perspective that all tracked satellites were physically converging on the static receiver simultaneously (see Figure 14). Indeed, verification of the satellite trajectories over the test area using a tracking satellites tool confirmed this thesis. A more rigorous hypothesis points to an uncorrected hardware clock drift within the mass-market oscillator. Because the WLS navigation filter natively absorbs common-mode timing errors into the user clock bias state, this hardware frequency drift maps directly into the observables, uniformly shifting all PRR derivatives downward and generating the illusion of a systemic convergence.

3 CONCLUSION

This study evaluated mass-market Android GNSS receiver resilience against environmental and adversarial interference. Severe multipath and NLoS reflections compromised accuracy by introducing substantial spatial variance. While C/N0-based filtering successfully stabilized the PVT solution against these artifacts, receivers remained highly vulnerable to intentional attacks. High-power RF interference desensitized the front-end, causing widespread tracking instabilities. Furthermore, algorithmic delay injections bypassed the WLS filter, instantaneously hijacking the navigation solution and exposing inherent susceptibilities to cyberspoofing.

A primary limitation remains the reliance on software-injected spoofing rather than over-the-air RF transmissions. Although this approach validates estimator failure and PRR divergences, it inherently bypasses hardware-level AGC variations essential for genuine interference detection. Future research should prioritize physical RF spoofing simulations to rigorously evaluate integrated AGC and C/N0 cross-checking countermeasures.

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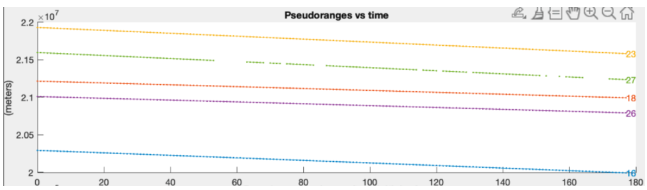


Figure 1

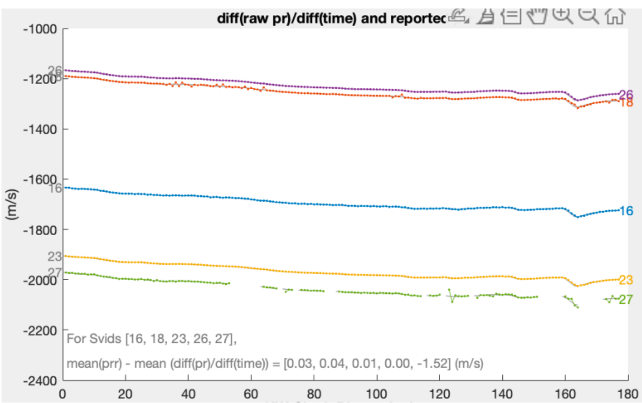
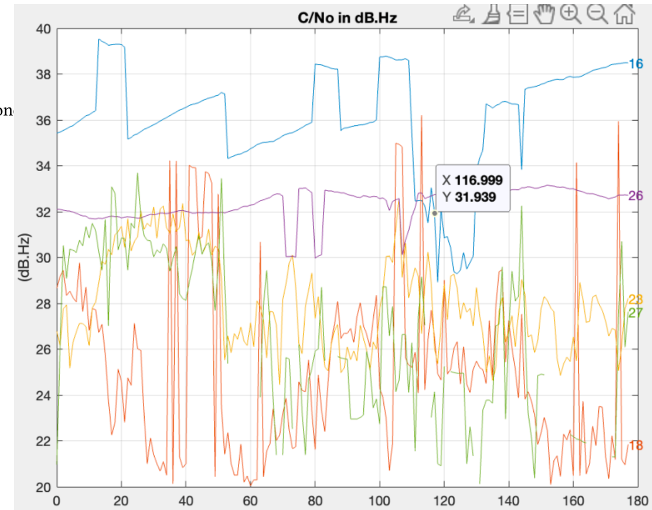


Figure 2

Table 1: Comparison of WLS Position Metrics

Parameter	Baseline	Filtered	Spoofed
Alt. Std. Dev. (m)	98.02	4.64	467.48
Alt. Maximum (m)	625.25	341.87	2141.39
Alt. Minimum (m)	303.37	299.18	-1058.98
Alt. Range (m)	321.88	42.69	3200.36
50% CEP (m)	44.60	4.48	122.22
Horiz. Std. Dev. (m)	66.23	4.64	191.21
East Variance (m^2)	4177.05	19.18	34309.82
North Variance (m^2)	209.25	2.39	2252.29
Max. Horiz. Error (m)	220.79	8.59	554.40



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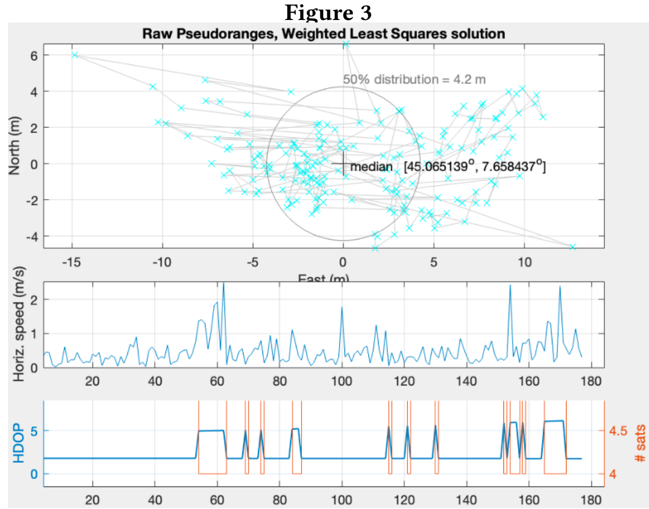

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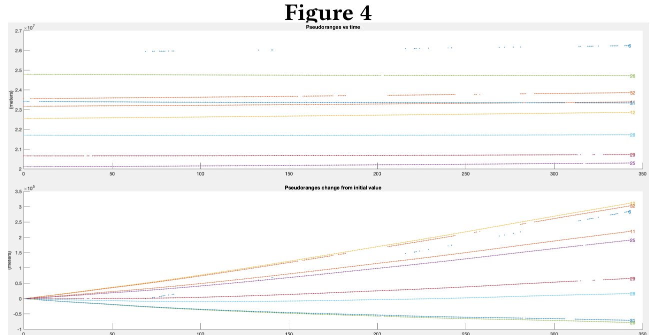
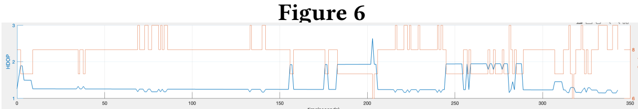


Figure 5

Figure 6



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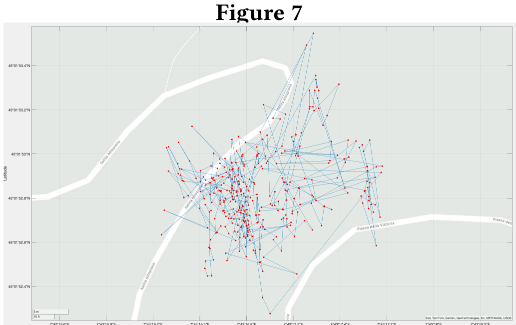

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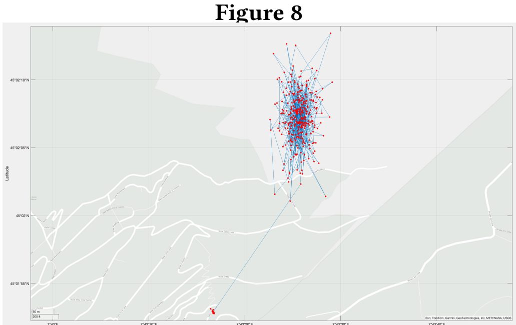

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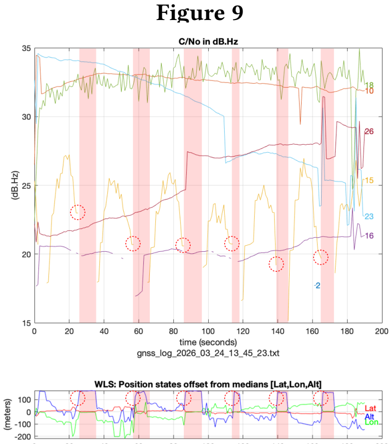
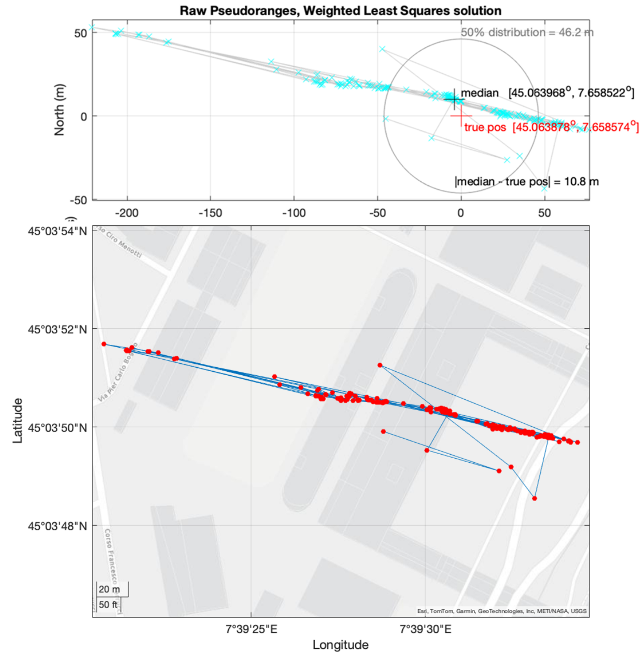


Figure 10



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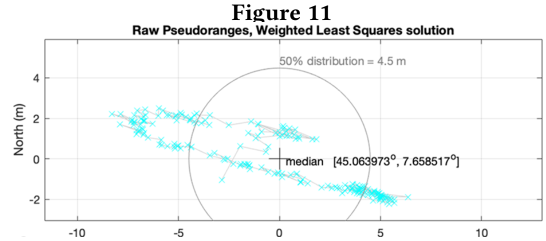

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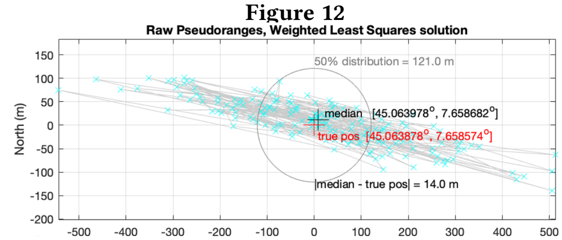

Figure 13

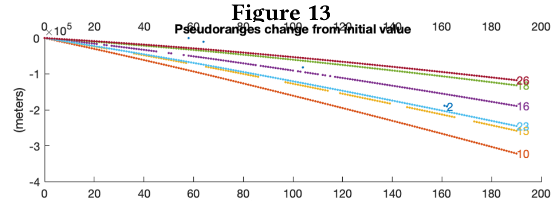

Figure 14